National University of Computer and Emerging Sciences



LABORATORY MANUAL

for

Data Warehousing & Business Intelligence Lab

(DL3004)

LAB 05 • Dimensional Modeling I: Grain, Facts, Dimensions & Keys

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Lab Manual 05

Objectives

By the end of this lab, students should be able to:

- Define a business process and state a precise dimensional grain before selecting tables.
- Distinguish facts from dimensions and transaction grain from snapshot/event grain.
- Differentiate natural/business keys from warehouse surrogate keys and justify when each is used.
- Classify measures as additive, semi-additive, or non-additive and avoid invalid aggregation.
- Design a sales transaction star schema from operational source extracts.
- Design a periodic snapshot fact table and a factless fact table for different analytical questions.
- Identify degenerate dimensions and decide when an attribute belongs in a dimension versus a fact.
- Defend dimensional-model design choices using business questions and grain statements.

Tools	PostgreSQL, pgAdmin 4, diagrams.net / dbdiagram.io
Datasets	Lab 05/06 source extracts + inventory snapshot + promotion events
CLO Mapping	CLO 2
Submission	SQL file + three dimensional-model diagrams + screenshots + modeling evidence document

PRE-LAB: Requirement

Pre-Lab Requirement
Download the Lab 05/06 dataset pack. Students should already understand primary keys, foreign keys, normalization, star schemas, and basic analytical SQL from Labs 03 and 04. No warehouse tables are created in this lab; the emphasis is dimensional design and modeling reasoning.

1. Business Scenario

CampusCart now wants a reusable enterprise data warehouse model. The source system stores customers, products, regions, orders, and order lines using operational business identifiers. Management wants sales analysis, inventory monitoring, and promotion-coverage analysis without forcing analysts to understand the operational schema.

Your job is to convert business questions into dimensional models. The central rule for this lab is: define the business process and grain first; facts, dimensions, keys, and measures come after the grain is fixed.

DESIGN PRINCIPLE: Grain first

A dimensional model is only reliable when every fact row has one unambiguous meaning. If the grain is vague, measures can be double-counted and dimensions can attach at the wrong level.

2. Supplied Files

File	Purpose	Important Fields
lab05_06_source_customers.csv	Operational customer master	customer_code, segment, region_code
lab05_06_source_products.csv	Operational product master	product_code, category, brand, list price
lab05_06_source_regions.csv	Region master	region_code, region_name, service_area
lab05_06_source_orders.csv	Order headers	order_id, order_date, customer_code, channel, status
lab05_06_source_order_items.csv	Order-line transactions	line_id, order_id, product_code, quantity, revenue measures
lab05_inventory_snapshot.csv	Month-end inventory observations	snapshot_date, product_code, region_code, on_hand_qty, inventory_value
lab05_promotion_events.csv	Promotion-product-region-channel events	event_id, event_date, promotion_name, product_code, region_code, channel

DATA NOTE

The customer and product extracts contain a small amount of duplicate/casing/whitespace noise. Do not clean it yet. Lab 06 will use staging transformations to handle those issues.

3. Profile the Source Before Modeling

Task 3.1 — Establish Source Grains

For each supplied source file, write one sentence describing what one row represents. Your answer must explicitly distinguish order headers from order lines.

Task 3.2 — Validate Candidate Business Keys

Using PostgreSQL temporary/raw tables or another reproducible method, compare total rows with distinct values of customer_code, product_code, region_code, order_id, and line_id. Record any source keys that are not unique and explain why a warehouse load cannot blindly assume source uniqueness.

Task 3.3 — Identify the Analytical Business Processes

From the supplied data, identify at least three distinct analytical business processes. At minimum include Sales Transactions, Inventory Snapshot, and Promotion Coverage/Occurrence. Do not combine them into one fact table.

4. Grain: The Contract of the Fact Table

Task 4.1 — Sales Transaction Grain

Complete a precise grain statement for a future FactSales table. Your statement must include the lowest-level transaction represented and must be precise enough that two developers would create the same fact table.

One row in FactSales represents _____.

Task 4.2 — Test the Grain Against Business Questions

For each question below, state whether the proposed sales grain can answer it correctly and explain why:

- Net revenue by month and category.
- Average selling price by product.
- Number of distinct orders by region.
- Current inventory on hand by product.
- Which promotion combinations existed even if they produced no sale?

Task 4.3 — Grain Error Challenge

A developer proposes one row per order but also wants product_code and product-level quantity in the same fact table. Explain the contradiction and provide two valid redesign options.

5. Facts, Dimensions, and Attribute Placement

Task 5.1 — Identify Dimensions

Design the conformed dimensions required for Sales: Date, Customer, Product, Region, and Channel. For each dimension, list its descriptive attributes and its natural/business key from the source.

Task 5.2 — Identify Facts and Degenerate Dimensions

Classify the following: quantity, unit_price_pkr, discount_pct, gross_revenue_pkr, discount_amount_pkr, net_revenue_pkr, order_id, and line_id. Decide which are measures, which are transaction identifiers, and whether order_id is a reasonable degenerate dimension.

CONCEPT: Degenerate dimension

A degenerate dimension is a business identifier stored directly in the fact table without a separate descriptive dimension table. Transaction numbers such as order_id are common examples when there are no meaningful descriptive attributes to place in a dimension.

6. Measure Additivity

Task 6.1 — Classify Sales Measures

Measure	Additive Across Rows?	Across Time?	Reason
quantity			
gross_revenue_pkr			
discount_amount_pkr			
net_revenue_pkr			
unit_price_pkr			
discount_pct			

Complete the table. For non-additive measures, state the safer aggregation (for example weighted average or ratio from additive components).

Task 6.2 — Inventory Snapshot Additivity

Using lab05_inventory_snapshot.csv, classify on_hand_qty and inventory_value_pkr. Explain why summing one month-end balance across products may be valid while summing the same product balance across multiple snapshot dates can be misleading.

Task 6.3 — Derived Ratio Challenge

Management wants Discount Rate = Total Discount / Gross Revenue. Explain why AVG(discount_pct) may not equal the correct overall discount rate. Write the correct aggregate expression conceptually or in SQL.

7. Natural Keys vs Surrogate Keys

Task 7.1 — Key Classification

Classify customer_code, product_code, region_code, order_id, and line_id as source/business identifiers. For each major dimension, propose a warehouse surrogate key name such as customer_key or product_key.

Task 7.2 — Why Surrogate Keys?

Give three warehouse-specific reasons to retain a surrogate key even when a natural key is currently unique. At least one reason must relate to history/versioning and one to independence from source-system changes.

Task 7.3 — Surrogate-Key Mapping Walkthrough

Assume source customer_code C0042 maps to customer_key 117 and product_code P0018 maps to product_key 36. Sketch how one order-line fact row would store the surrogate keys while still retaining order_id as a degenerate identifier.

8. Design the Sales Star Schema

Task 8.1 — Create the Diagram

Using diagrams.net or dbdiagram.io, draw FactSales in the center connected directly to DimDate, DimCustomer, DimProduct, DimRegion, and DimChannel. Include surrogate PKs in dimensions and corresponding FKs in the fact. Retain source business keys inside the dimensions for traceability.

Task 8.2 — Annotate the Model

Your diagram must show: fact grain, PKs, FKs, business keys, degenerate order_id, analytical measures, and at least two hierarchies: Year → Quarter → Month → Date and Category → Brand → Product.

Task 8.3 — Dimension Placement Challenge

Decide where payment_method and order_status should live for this model. You may choose a small dimension, a junk dimension, or store a constrained attribute in the fact. Justify your choice based on reuse, cardinality, and reporting needs.

9. Design Two Additional Fact Models

9.1 Periodic Snapshot — Inventory

Task 9.1

Design FactInventorySnapshot using lab05_inventory_snapshot.csv. Define its grain and connect appropriate dimensions. Identify which balances are semi-additive and state the correct interpretation across time.

9.2 Factless Fact — Promotion Coverage

Task 9.2

Design a factless fact table from lab05_promotion_events.csv. The table should record that a promotion-product-region-channel combination existed on a date even if no numeric measure is stored.

Task 9.3 — Business Questions

Write two questions that FactInventorySnapshot can answer better than FactSales and two questions that the factless promotion fact can answer without requiring a numeric measure.

10. Applied Modeling Challenge

For each scenario, state: business process, grain, dimensions, facts/measures, key strategy, and one likely modeling error to avoid.

Scenario	Required Modeling Decision
A — Sales Returns	Would you model returned lines in the same sales fact using signed measures/status, or as a separate returns process? Defend one choice.
B — Daily Account Balance	A balance is captured once per customer per day. Explain additivity across customers versus across days.
C — Student Event Attendance	Only attendance occurrence is required. Determine whether a factless fact is appropriate.
D — Multi-Source Product IDs	Two source systems can both use product_code P100 for different products. Explain why a warehouse surrogate key becomes essential.

11. Submission Requirements

- Lab05_<RollNo>.sql containing source profiling and key/grain validation queries.
- One sales star-schema diagram with grain, facts, dimensions, natural keys, surrogate keys, PK/FK links, and hierarchies.
- One inventory periodic-snapshot diagram and one factless promotion diagram.
- A short evidence document containing measure-additivity answers, key reasoning, and Applied Modeling Challenge A–D.
- Screenshots of source profiling results.

QUALITY CHECK: Before submission

Every fact table must have exactly one explicit grain statement. If a dimension or measure cannot be justified at that grain, redesign before submission.